TriSense AQMS — Monitoring Criteria, Averaging Methodology and TEMS Alarm Configuration

This document sets out the technical basis for configuring TEMS across the five TriSense AQMS stations, covering sensor ranges, alarm thresholds, averaging methodology and data-validity logic. Thresholds referenced here are operational screening and action criteria for TEMS configuration, not formally established regulatory limits unless stated otherwise.

1. Sensor Measurement Ranges, Detection Limits and Units

Parameter	Sensor Principle	Measurement Range	Detection Limit	Units	Basis / Notes
H_2S	Electrochemical	0–100 ppm; linear to 200 ppb	±2 ppb	ppm / ppb	—
NH_3	Electrochemical	0–100 ppm; overgas tolerance to 200 ppm	< ±10 ppm equivalent	ppm	—
VOC	Photoionization detector (PID)	0–40 ppm	1 ppb	ppm / ppb	—
Wind speed / direction	Ultrasonic or equivalent	0–60 m/s (116 knots); 0–360°	N/A	m/s, °	—

2. Data Recording and Transmission Intervals

Each channel is sampled internally at a rate sufficient to characterise sensor response and support stable averaging, wind at a 1-second scan, gas channels every 2 minutes, with raw samples buffered on-station rather than individually transmitted. Wind is reported as a 10-minute rolling mean, aligned with WMO convention. Each gas channel is reported as a 10-minute rolling average, computed only from valid records as defined in Section 4, and this is the value that drives the Warning, Action and Critical alarm logic in Section 3. Longer averaging windows, where required for external reporting, are a server-side TEMS configuration and are not fixed by station firmware. TEMS transmits every 10 minutes per Section 7's data package; there is no separate duty-cycled or hourly-batch transmission mode.

Convention	Sampling Period	Averaging Period	Averaging Type	Transmission Cadence	Remarks
Surface wind observation	1-second scan	10-minute mean	Rolling arithmetic mean of valid records	10 min	WMO supports the 10-minute mean convention.
H2S ambient observation	2 minutes	10-minute mean	Rolling arithmetic mean of valid records	10 min	Station-level average.

Convention	Sampling Period	Averaging Period	Averaging Type	Transmission Cadence	Remarks
NH3 ambient observation	2 minutes	10-minute mean	Rolling arithmetic mean of valid records	10 min	Station-level average.
VOC ambient observation	2 minutes	10-minute mean	Rolling arithmetic mean of valid records	10 min	Station-level average.

3. Warning, Action and Critical Alarm Thresholds

Formal numerical trigger values are not defined at this stage, as site-specific ambient conditions and variability have not yet been established. An initial 72-hour monitoring period, followed by a 30-day baseline review, will be used to develop a representative site baseline. Warning, Action and Critical trigger levels will subsequently be calculated using the adopted statistical methodology described below.

Station Alarm Floor: A minimum trigger concentration established during commissioning to prevent alarms from being generated by sensor noise, measurement uncertainty, or values near the analytical detection limit. The alarm floor shall be set by TRIDEL based on sensor performance characteristics, commissioning observations, and the established site baseline, and may be reviewed during the 30-day baseline assessment period. The alarm floor is not a regulatory limit and serves only as a minimum operational threshold for alarm generation.

Level	Trigger Methodology	Persistence / Confirmation	TEMS Response
Warning	Maximum of station alarm floor and B + 2S	Two consecutive valid 10-minute averages, or one valid 30-minute average	Create Warning event and transmit immediately
Action	Maximum of station alarm floor and B + 3S	Two consecutive valid 10-minute averages, one valid 30-minute average, or qualified rapid rise	Create Action event and transmit immediately
Critical	Maximum of station alarm floor and B + 4S	One valid 10-minute average; confirmation does not delay notification	Create Critical event and transmit immediately

4. Data-Validity and Completeness Requirements

A record is valid when its timestamp is credible, channel status is normal, the value is within range, required compensation inputs are available and the station is not in warm-up, calibration or maintenance mode. Invalid records remain stored with a reason code but are excluded from averages and environmental alarm calculations. Missing data remains missing and is never substituted with zero or carried forward from the last value. Station clocks are synchronised to a common time zone; a delayed packet retains both its original measurement time and its receipt time, so late data is not misattributed to the wrong period.

Calculated Metric	Expected Records	Minimum Valid Records	Treatment Below Minimum
10-minute gas average from 2-minute records	5	4	Flag invalid; no environmental alarm from this metric
Minimum data-completeness threshold for a valid reporting period	—	75% of expected valid records	Reporting period flagged incomplete and reported as such, rather than withheld

5. Alarm Notification and Acknowledgement

Event	Notification	Proposed Acknowledgement Target	Initial Review
Warning	Immediate TEMS notification	Within 4 hours	Review trend, quality, wind and nearby station behaviour
Action	Immediate notification and escalation	Within 2 hours	Start operational investigation and confirm event status
Critical	Immediate urgent escalation	Within 30 minutes	Urgent review under the DM escalation procedure

Acknowledgement targets are indicative operational controls and remain subject to Dubai Municipality approval and the agreed monitoring coverage.

6. Exceedance and Red-Flag Logic

TEMS supports alert acknowledgement, operator notes, an audit trail and evaluation of alert rules against incoming data, with DM administrator, DM operator and TRIDEL service roles distinguished. The 72-hour and a further 30-day baseline processes identify anomalies, baseline shift, abnormal noise, flat-line behaviour, abrupt shifts, and consistency with local wind and operations, which trigger a controlled spare-sensor drift check rather than an operational alarm. The statistical trigger values will be finalised following completion of the baseline assessment; however, the alert generation, persistence and escalation logic described below will remain applicable.

Item	Proposed TEMS Logic
Environmental exceedance	Apply Warning, Action and Critical persistence rules only to valid averages. Record the station, parameter, value, threshold, averaging period, quality status and event time.
Notification recipient(s) on exceedance	Primary: DM Operator, the role already defined in TEMS, receives and acknowledges the alert in-app.
Rapid rise	Create an Action event when the current valid 10-minute average is at least twice the median of the preceding three valid 10-minute averages and is above the Warning threshold.
Alarm clearance	Warning clears after three consecutive valid 10-minute values below Warning. Action and Critical also require operator acknowledgement. Retain the event history.

Item	Proposed TEMS Logic
Communication loss	Create a technical alert when no expected 10-minute packet is received within 30 minutes of the previous successful receipt (proposed: 3× cadence,). Backfill stored records when communication returns.
Technical red flags	Sensor fault or out-of-range status; stale or duplicate timestamp; unexplained flat line longer than 60 minutes; confirmed time error; power or enclosure limit violation.
Alert routing	Technical faults are routed to Tridel support and are not labelled environmental exceedances. Environmental alarms are routed to configured DM recipients.

7. TEMS Data Package

Content	Minimum Data Transmitted Every 10 Minutes
Measurements	All stored two-minute H2S, NH3 and TVOC records for the completed 10-minute period, together with wind statistics
Summary	10-minute average, minimum and maximum
Validity	Expected count, valid count, completeness percentage, quality flag and reason code
Station status	Station ID, timestamps, configuration version, sensor/logger status, power status and communication status

8. Recognized Technical References and Assumptions

Reference Basis:

No.	Reference
1	U.S. EPA, A Guide to Siting and Installing Air Sensors, updated 7 January 2026.
2	U.S. EPA, Enhanced Air Sensor Guidebook (2022) — network design, QA/QC and data-management guidance.
3	U.S. EPA, Quality Assurance for Air Sensors — project planning, maintenance, troubleshooting and routine QC guidance, updated 11 March 2026.
4	U.S. EPA, Frequent Questions about Landfill Gas — composition and NMOC formation.
5	ATSDR, Landfill Gas Primer, Chapter 2 — used only for foundational landfill-gas composition context, not as a monitoring standard.
6	UAE Cabinet Resolution of 2023 Concerning the Technical Regulations for Air Quality Measurement — calibration and network/instrument controls.